AI Image Generator

9. Admin Panel

Version 2.0
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Platform: NVIDIA GeForce RTX 3090, Windows 11, Python 3.12

Z-Image Turbo, Stable Diffusion XL and its best fine-tunes, FLUX.1, LoRAs and Real-ESRGAN, served locally with Gradio in the Frazile AI studio UI, with per-image GPU telemetry recorded in SQLite and a live Admin panel.

9. Admin Panel

The Admin tab is a live view of the machine running the studio. Every number is sampled on demand by sysinfo.py; nothing is stored. A refresh costs about 0.7 s (3 s the first time, when folder sizes are computed). The tab refreshes when opened, on the Refresh button, and every 5 s when auto-refresh is ticked.

9.1 Tiles

Tile	Value	Meter or state	Source
GPU load	utilisation %, SM clock, P-state	meter	nvidia-smi
VRAM used	GiB used of total; torch allocation	meter of used %	nvidia-smi, torch
GPU temperature	°C and fan %	state: OK under 70, Warm under 83, Hot under 90, Critical above	nvidia-smi
GPU power	W of limit, % of limit	meter	nvidia-smi
CPU	usage %, threads, frequency	per-core bar strip	psutil
Memory	GB used of total	meter	psutil
System disk	GB free, % used	meter	psutil
Network	download and upload rate, established connections	-	psutil, two samples
App uptime	uptime, PID, resident memory	-	psutil

Meters use the pink accent and switch to the amber warning colour above 85 % and the red critical colour above 95 %. The temperature state always carries an icon and a word, never a colour alone.

9.2 Sections

Section	Fields
GPU	Card, VRAM, compute capability and SM count; driver, CUDA, cuDNN; current and maximum SM and memory clocks; GPU and memory-bus utilisation; VRAM from nvidia-smi and from torch (allocated, reserved, session peak); PCIe generation and width; throttle reasons decoded from the nvidia-smi bitmask (idle, power cap, thermal slowdown and so on); VBIOS and PCI bus; loaded model and sampler; processes on the GPU.
CPU	Processor name, architecture, physical and logical cores, current and maximum frequency, total and per-core usage, 1/5/15-minute load averages, process and thread counts, boot time and uptime.
Memory and storage	RAM used, available and total; swap or page file; disk reads and writes since boot; outputs folder size and PNG count; models folder; Hugging Face cache size and path; database size; every partition with file system, total, used, free and percentage.
Network	Hostname and primary IP; current throughput (rates need two samples); bytes and packets since boot; errors and drops; established connections and listening ports; every interface with state, IPv4, MAC, link speed, sent and received; optional reachability of huggingface.co, github.com and pypi.org in milliseconds.
Power	GPU draw against the configured limit with percentage; default and maximum limits; performance state and fan; the active Windows power plan; battery if present; estimated GPU energy of all recorded generations in Wh.
Software and app	OS, Python, torch, CUDA, cuDNN, diffusers, transformers, gradio versions; whether the process runs in Docker; app PID, start time, uptime, resident memory and threads; working directory; generation totals from the database.

9.3 Reading the numbers

- Idle readings on the RTX 3090 are about 45 W, P8, SM clock 210 MHz, PCIe gen 1 x1 and throttle reason "GPU idle". Under load expect P0, 1.7 to 1.9 GHz, PCIe gen 3 or 4 x16 and 350 to 390 W.
- nvidia-smi reports allocator reservation as used VRAM; the torch allocated figure is the live tensor total.

- Process names on the GPU may show "Insufficient Permissions" for processes of other users or the system.
- The network rate tiles show 0 on the first sample after start; every later refresh shows the rate since the previous one.
- Inside Docker the CPU, memory and network sections describe the container, while GPU and power come from the host driver.